[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINE, CYLINDRICAL, EXTERNAL, CENTER-TYPE, COMPUTER NUMERICAL CONTROL (CNC)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE This commercial item description (CID) covers computer numerical control (CNC) center-type external cylindrical grinding machines. The grinding machines covered by this CID are intended for use in operations where multiple workpieces are ground from workpiece programs to close tolerances for outside diameter, straightness, tapers, steps, and faces.

2. CLASSIFICATION The grinding machines shall be of the following types, swings and sizes. The type, swing and size to be furnished shall be as specified (see 7.2(b)).

Type I – Traversing worktable Z-axis, traversing wheelhead infeed X-axis
Type II - Traversing wheelhead, Z- and X-axes (longitudinal and infeed); fixed worktable

Swing - (see table I)

Size - (see table I)

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

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TABLE I. Type I and type II machine swing and sizes.

Swing, inches	Sizes - center-to-center distance, inches																			
	7	12	15	23	25	32	39	59	72	96	118	137	157	177	196	236	275	315	354	393
7	XZ	X																		
9	Z		XZ		XZ															
10			X		X		X													
11			XZ	XZ	Z		X	X												
12				Z			Z	Z	Z											
13		X	X		X		X	X												
16		X	X	X	X	X	X	X												
17				X			X	X												
19				Z			Z	Z	Z	Z	Z	Z	Z	Z	Z	Z				
20						Z	Z													
23							Z	Z	Z	Z	Z	Z	Z	Z	Z	Z	Z			
24								Z												
26									Z											
30										Z										
31							Z		Z		Z		Z		Z	Z	Z			
35									Z		Z		Z		Z	Z	Z	Z	Z	Z

Note: X denotes availability of swing and sizes for type I machines. Z denotes availability of swing and sizes for type II machine.

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Identification of axes. The grinding machine shall be provided with X- and Z-axes of linear motion in accordance with American National Standards Institute/Electronic Industries Association (ANSI/EIA) 267-C.

3.2 Components. The grinding machine shall include, as a minimum, the following components:

3.2.1 Base. The base shall support the other elements of the machine and shall incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base shall be provided with a means for leveling and anchoring the machine. The base and integral machineways shall be constructed to prevent the center of gravity of the wheelhead assembly, workhead, or tailstock from extending beyond the periphery of the base or past the end of the supporting machineways.

3.2.1.1 Machineways. Unless otherwise specified (see 7.2(c)), machineways shall be of the design considered by the manufacturer as standard for the machine. All machineways shall have a protective means to keep swarf and other debris from bearing surfaces.

3.2.2 Wheelhead assembly. The wheelhead assembly shall consist of a grinding wheel, grinding wheel spindle, drive mechanism, wheel guard, and assembly housing mounted on a slide table. The wheel spindle shall be mounted on antifriction bearings designed to limit radial and axial play. The bearings shall be continuously lubricated or permanently lubricated and sealed for life. The axis of the grinding wheel spindle shall be perpendicular to the X-axis (infeed).

3.2.2.1 Wheelhead slide table. The slide table on a type I machine shall traverse (infeed) on the X-axis. The slide table on a type II machine shall be a compound slide that traverses on the X- and Z-axes. The X-axis shall be perpendicular (referred to as 0 degrees or straight) to the Z-axis. When an angular wheelhead is required, the X-axis shall be offset 30 degrees or adjustable over an angular range as specified (see 7.2(d)).

3.2.3 Workhead. The workhead assembly shall be longitudinally adjustable along its mounting table or machineways with a means for positive locking. The workhead shall be configured for dead- or live-center grinding as specified (see 7.2(e)). The workhead shall be provided with an internal taper to accommodate replaceable work centers. The workhead spindle shall be mounted on antifriction or low-friction bearings that are automatically lubricated or sealed for life and designed to minimize axial and radial runout.

3.2.3.1 Dead-center. A drive plate (complete with drive pin) that revolves around the work center shall be provided. The bearings for the drive plate shall be protected from intrusion of grinding swarf or other foreign materials, automatically lubricated or sealed for life, and designed to minimize generation or transmission of vibration to the workpiece.

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3.2.3.2 Live-center. When a live-center configuration is provided, the workhead spindle shall be configured with an internal taper designed for the use of replaceable centers, collets, or chucks.

3.2.4 Tailstock. The tailstock assembly shall be longitudinally adjustable along its mounting table or machineways with a means for positive locking. The tailstock spindle shall be longitudinally extendible utilizing a handwheel, lever, or other means. The centers for the tailstock spindle shall be interchangeable with, or at least adaptable to, the spindle of the workhead. The spindle shall have a means to adjust workpiece center contact pressure.

3.2.5 Worktable. Unless otherwise specified (see 7.2(f)), a worktable shall be provided to support a swivel table and/or the workhead and tailstock.

3.2.6 Swivel table. When required, a swivel table shall be provided for mounting the workhead and tailstock and have an angular range of swivel as specified (see 7.2(g)). The swivel table shall be provided with a locking means to provide a stable platform for grinding tapers or offset contours.

3.2.7 CNC system. The CNC system shall be based on either a personal computer (PC) open architecture or other software-programmable microprocessor design. The CNC system shall be industrial grade in accordance with accepted industry practices and shall meet Government requirements as specified (see 7.2(h)).

3.2.7.1 Hardware features. Unless otherwise specified (see 7.2(i)), the CNC system shall include, as a minimum, the following hardware:

a. One or more 32-bit microprocessors with math coprocessor.
b. 250-MB rewritable nonvolatile memory.
c. 3.5-inch high density floppy disk drive.
d. DNC link capability.
e. High-visibility monitor (minimum screen 9-inch diagonal).
f. Manual data input (MDI) device.
g. Dual EIA RS-232-E communication ports.

3.2.7.2 Operator controls. The operator controls, display panel, alarms, and indicators shall be located at the operator's workstation and shall be convenient to and in full view of the operator during all staging and operation. Unless otherwise specified (see 7.2(j)), the operator controls shall provide the following capabilities:

a. Emergency shutdown.
b. Control on.
c. Program start.
d. Program pause.
e. Program stop.
f. Display of operator alerts and alarms.
g. Display of active program progress.
h. Manual override of active program.

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i. Axis select and increment.
j. Rapid positioning.
k. Variable jog.
l. Variable infeed.

3.2.7.3 Software features. Unless otherwise specified (see 7.2(k)), the CNC system shall include, as a minimum, the following software programming and control features:

a. Simultaneous control of all CNC axes.
b. Linear, circular, parabolic, and helical interpolation.
c. Inch/metric switchable programming.
d. Decimal point programming.
e. Workpiece graphic display rotation.
f. Tool path graphic display.
g. Standard EIA 494-B codes compatibility.
h. Dimensional absolute/incremental programming.
i. Cartesian/polar coordinate programming.

3.2.7.4 Operating modes. Unless otherwise specified (see 7.2(l)), the CNC system shall include, as a minimum, the following operating modes:

a. Automatic operation.
b. Manual operation.
c. Simulated machine operation.
d. Manual data input (MDI).
e. Setup.
f. Learn.
g. Stored program editing.
h. Background editing/split screen program preview.
i. Machine diagnostic/maintenance/monitoring.

3.2.7.5 Axis management. Unless otherwise specified (see 7.2(m)), the CNC system shall include, as a minimum, the following axis management features:

a. Wheel wear compensation.
b. Backlash compensation.
c. Workzone/motion collision/limit detection.
d. Repeatable error correction/compensation.
e. Machine zero reference reset.

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3.2.8 Drive mechanisms. All drive mechanisms shall be capable of continuously operating the machine at maximum speed, grinding load, and capacity. All gears, pinions, shafts, and coupling devices for spindle and axis drive trains shall be manufactured of suitable materials and treated to assure continued accuracy and alignment without failure for the expected life of the grinding machine. All belt-driven systems shall be provided with belt-size compensation and tensioning devices to assure proper function and long life.

3.2.9 Grinding wheel dresser. Unless otherwise specified (see 7.2(n)), a grinding wheel dresser with at least a one-carat diamond shall be provided to true the face of the grinding wheel. The dresser shall be capable of truing the grinding wheel to its full usable depth. Dressing shall not require removal of the workpiece.

3.2.10 Dust extractor. When specified (see 7.2(o)), a dust extractor shall be provided. When provided, the dust extractor and the coolant system shall be interlocked to allow separate operation of either system and prevent simultaneous operation of both systems.

3.2.11 Coolant system. Unless otherwise specified (see 7.2(p)), a flood type coolant system shall be provided with all necessary equipment and sufficient reservoir capacity to permit a continuous full flow of coolant. The system shall be provided with a means for draining and cleaning and shall include a baffle or strainer that is easily removable for cleaning. A means shall be provided to permit the operator to direct and control the flow of coolant through the entire work area.

3.2.12 Electrical system. Unless otherwise specified (see 7.2(q)), the electrical system shall conform to the National Fire Protection Association (ANSI/NFPA)-79. The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(r)). Unless otherwise specified (see 7.2(s)), the control system, shall be provided with a separate 120-volt, single-phase, 60-hertz circuit from the machine main power.

3.2.13 Electric motors. The grinding machines drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1. Unless otherwise specified (see 7.2(t)), the motors shall have open dripproof enclosures.

3.2.14 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(u)).

3.3 Performance. Unless otherwise specified (see 7.2(v)), the grinding machine shall meet the alignment tolerances shown in table II and shall produce workpieces that meet the performance requirement shown in table III.

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TABLE II. Alignment tolerances.

Parameter	Allowable deviation
Bed straightness in longitudinal direction	±0.0001 inch per foot
Bed flat or level in infeed direction, no twist permitted	±0.0001 inch per foot
Axis of workhead and tailstock parallel with table movement	±0.0001 inch per foot
Axis of wheelhead parallel with table movement in vertical and horizontal plane	±0.0002 inch per foot
Wheelhead spindle axial slip and radial runout	0.0002 inch TIR[1]
Wheelhead spindle flange axial runout	0.0002 inch TIR
Axis of workhead parallel with table movement in vertical and horizontal plane	±0.0001 inch per foot
Workhead spindle radial runout	0.0001 inch TIR
Workhead spindle axial slip at two points displace by 180 degrees	0.0001 inch TIR
Workhead spindle flange axial runout	0.0001 inch TIR
Tailstock sleeve parallel with table movement in vertical and horizontal plane	0.0005 inch per foot TIR

[1] TIR is total Indicator Reading

TABLE III. Performance requirements.

Parameter	Allowable deviation
Dimensions	0.0001 inch TIR
Roundness	0.0001 inch TIR
Straightness	±0.0001 inch per foot
Finish	5 microinches Rt[1]

[1] Rt is roughness total as defined in American Society of Mechanical Engineers (ASME) B46.1.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(w)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.9. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(x)). Regardless of the measurement system used, all measuring and indicating devices on the machine

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shall be graduated in the same system. When specified (see 7.2(y)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(z)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(aa)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The grinding machine shall be inspected to determine compliance with all requirements specified in this CID.

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5.3 Testing. Unless otherwise specified (see 7.2(bb)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(cc)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with certification traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(dd)). Unless otherwise specified (see 7.2(ee)), the grinding machine shall be tested in the environmental conditions prevalent at the manufacturer's own test facility.

5.3.2 Operational test. The grinding machine shall be operated in accordance with the manufacturer's standard operating procedures. The grinding machine shall be operated at no load for not less than two hours. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Alignment tests. The grinding machine shall be checked for alignment before and after the performance tests and shall conform to the alignment requirements shown in table II. Changes in alignment shall be cause for rejection.

5.3.4 Performance tests. Performance tests shall be conducted in accordance with the manufacturers recommendations and procedures. Unless otherwise specified (see 7.2(ff)), a test workpiece (see figure 1) shall be ground from a solid-steel cylinder that is at least half the center-to-center capacity of the machine in length (L) and half the swing capacity in diameter (D). The workpiece dimensions shall be stipulated by the inspector prior to the test. The curved shoulder and the tapered step shall be ground with a square-faced grinding wheel using linear and circular interpolation. After verification of the accuracy of the curve and taper, the wheel may be changed or dressed for finish grinding. The test results shall be compared to the performance requirements shown in table III. Following completion of the workpiece, the alignment of the machine shall be verified again and the results compared with those recorded during the operational test.

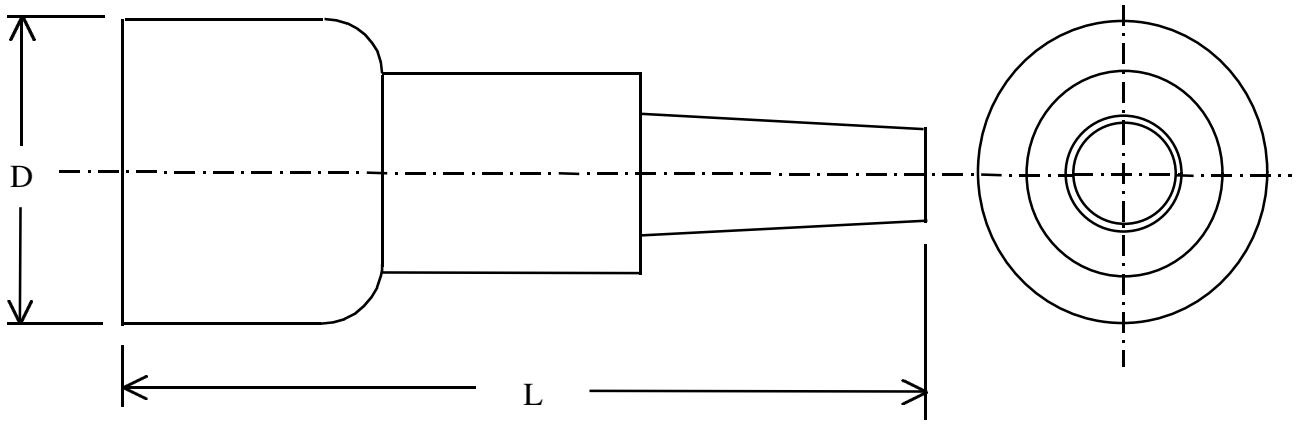


FIGURE 1. Performance test workpiece.

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5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(gg)).

5.4 Acceptance. Unless otherwise specified (see 7.2(hh)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared to the requirements shown in tables II and III. Failure of the machine to meet these requirements shall be cause for rejection.

6. PACKAGING. Preservation, packing, and marking shall be as specified (see 7.2(ii)).

7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B11.9	Safety Requirements for the Construction, Care, and Use of Grinding Machines

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

American Society of Mechanical Engineers (ASME)

ASME B46.1	Classification and Designation of Surface Qualities

Applications for copies should be sent to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

Occupational Safety and Health Association (OSHA)

OSHA 29 CFR PART 1910	Occupational Safety and Health Standards for General Industry

Applications for copies should be sent to the U. S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington DC 20210.

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Electronic Industries Association (EIA)

ANSI/EIA 267-C	Axis and Motion Nomenclature for Numerically Controlled Machines
ANSI/EIA RS 232-E	Interface between Data Terminal Equipment and Data Circuit terminating Equipment Employing Serial Binary Data Interchange
ANSI/EIA 494-B	32 Bit Binary CL (BCL) and 7 Bit ASCII CL (ACL) Exchange Input Format for Numerically Controlled Machines

Applications for copies should be sent to the Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3834.

National Electrical Manufacturers Association (NEMA)

NEMA MG-1	Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

ANSI/NFPA 79	Electrical Standards for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

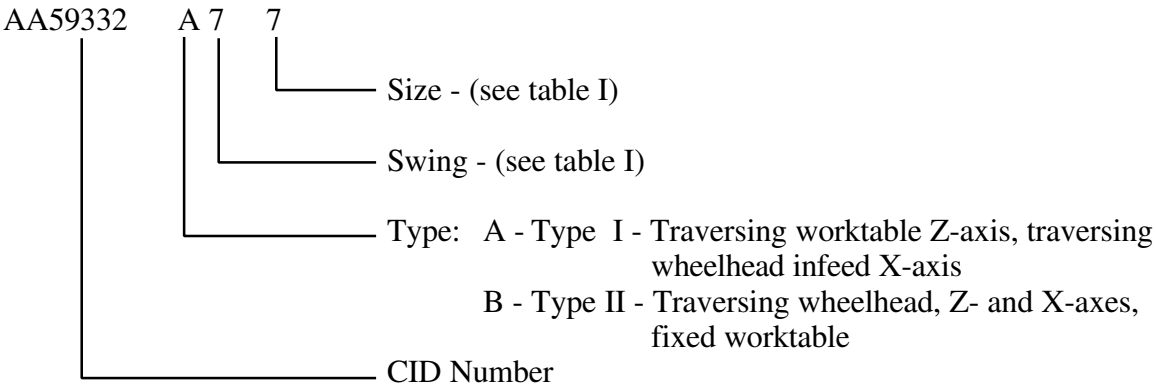
7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type, swing, and size required (see 2).
c. Machineways, if different (see 3.2.1.1).
d. Angular wheelhead, as specified (see 3.2.2.1).
e. Live/dead centers, as specified (see 3.2.3).
f. Worktable, if different (see 3.2.5).
g. Swivel table, as specified (see 3.2.6).
h. CNC system, as specified (see 3.2.7).
i. CNC system hardware, if different (see 3.2.7.1).
j. Operator controls, if different (see 3.2.7.2).
k. CNC software programming and control, if different (see 3.2.7.3).
l. CNC operating modes, if different (see 3.2.7.4).
m. CNC axis management, if different (see 3.2.7.5).
n. Grinding wheel dresser, if different (see 3.2.9).
o. Dust extractor, if required (see 3.2.10).
p. Coolant system, if different (see 3.2.11).
q. Electrical system specification, if different (see 3.2.12).
r. Machine voltage, as specified (see 3.2.12).

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s. Control system voltage, if different (see 3.2.12).
t. Motor enclosures, if different (see 3.2.13).
u. Optional accessories, as specified (see 3.2.14).
v. Performance, if different (see 3.3).
w. Machine dimensions and weight, if required (see 3.4).
x. System of units required (see 3.6).
y. Dual system of units, if required (see 3.6).
z. Nameplate, if different (see 3.7).
aa. Lubrication plate or chart, if required (see 3.8).
bb. Testing, if different (see 5.3).
cc. Test responsibility, if different (see 5.3.1).
dd. Certification of calibration, as specified (see 5.3.1).
ee. Test environmental conditions, if different (see 5.3.1).
ff. Performance test, if different (see 5.3.4).
gg. Optional tests, as specified (see 5.3.5).
hh. Acceptance, if different (see 5.4).
ii. Packaging requirements (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodians
Air Force - 99
Navy - SH

Reviewers
Air Force - 84
Navy - AS, OS

PREPARING ACTIVITY
DLA - GS

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